

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Sixth Semester of B. Tech (EC) Examination****May 2018****EC310.01/EC310 Optical Communication****Date: 05.05.2018, Saturday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION – I****Q - 1 Answer the following. [07]**

- 1) Justify the statement 04
  - a. “Skew ray provides more secure communication”
  - b. “Halide Glass fiber offers very low attenuation, but still not prefer for long distance communication”
- 2) What do you mean by dispersion? Describe 03
  - a. Chromatic dispersion
  - b. Polarization mode dispersion

**Q – 2.a** A step index multimode fiber with NA of 0.2 supports approx.1000 modes at an 850nm wavelength. Find the core diameter. **[04]****Q – 2.b Answer any two questions. [10]**

- 1) Define Bending loss and also explain types of bending loss occur in single mode fiber.
- 2) Discuss the mechanical misalignment in fiber optic cable with necessary sketch.
- 3) Discuss the advantage of optical communication in detail.

**Q - 3 Answer the following.**

- a. Discuss the different lensing schemes for coupling improvement. [05]
- b.
  - 1) Describe with a neat and labeled sketch the construction of edge emitting LED. [05]
  - 2) Explain three key transition processes involved in laser action. [04]

**OR**

- 1) Explain Distributed Feedback LASER. [05]
- 2) Explain modulation of LED. [04]

**SECTION – II****Q - 4 Answer the question below. [07]**

- a. One of the following is not a solid-state optical detector [01]  
 a. APD    b. PIN    c. PMT    d. Photo Transistor
- b. Define impact ionization. [01]
- c. DWDM stands for \_\_\_\_\_. [01]
- d. In an optical fiber, the concept of Numerical aperture is applicable in describing the ability of \_\_\_\_\_. [01]  
 a. Light Collection  
 b. Light Scattering  
 c. Light Dispersion  
 d. Light Polarization
- e. What is intermodal Dispersion? [01]
- f. Why we need to connect photodiode in reverse bias? [02]

**Q – 5.a Explain principal of operation, the structure and the characteristic of PIN Photo Diode. [05]****OR****Q – 5.a Draw and explain the structure and operation of InGaAs APD. [05]****Q – 5.b Which are the key features of WDM? [04]****Q – 5.c Write a short note on photo detector noise. [05]****OR****Q – 5.c Discuss optical power loss model for point to point link with necessary diagram. [05]****Q – 6.a A 2\*2 biconical tapered fiber coupler has an input optical power level of  $P_o=200\mu W$ . The output power at the other three ports are  $P_1=90\mu W$ ,  $P_2=85\mu W$ ,  $P_3=6.3nW$ . Find out coupling ratio, excess loss, insertion loss (from port 0to1 & 0to2) and crosstalk. [05]****Q – 6.b Explain SONET frame structure Give the difference between SONET and SDH. [05]****Q – 6.c Which factors affect the Rise Time Budget? In One optical communication link have the photo diode which required receiver input signal of -42dbm and optical source that can transmit -13dbm average optical power level in to the fiber fly head with 50 $\mu m$  core diameter. Assume that 1db loss occur when the fiber fly head is connected to the [04]**

cable and another 1db connector loss. Link also includes the 6db system margin. Find the possible transmission distant with  $\alpha_f = 3.5\text{db/km}$

**OR**

- Q – 6.a** Draw the schematic diagram of typical optical receiver and also explain different type of preamplifier in short. **[04]**
- Q – 6.b** Write a short note on 2 Fiber UPSR. **[05]**
- Q – 6.c** Explain 2 x 2 fiber coupler. What are the different losses associated with it? **[05]**

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